

Build4All Product Inventory and Coupon Policy

Controlled knowledge source for multilingual Text to SQL retrieval

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Authoritative schema	build4all_research
Status	Controlled research source
Scope	Products, categories, stock, prices, and coupon validity

Purpose and authority

This document defines product, inventory, category, price, and coupon meanings for the frozen build4all_research PostgreSQL schema. It is the authoritative controlled source for retrieval and must remain aligned with the benchmark gold queries.

Product fields and status

Each product is identified by products.product_id. The human-readable product fields are products.sku and products.name. Product catalog status is stored in products.status. Stock quantity is stored in products.stock. Catalog status and stock quantity are separate concepts and must not be substituted for each other.

Authoritative product and inventory rules

Product count by catalog status

Count every product and group by products.status. Do not apply an active-status or stock filter because the purpose of this metric is to show every catalog-status category.

Active products with fewer than ten units

Return products whose products.status is active and whose products.stock is strictly less than 10. Zero stock is included because zero is fewer than ten. Return sku, name, and stock when these business fields are requested. Order by stock and then sku for reproducibility.

Active inventory value

Active inventory value is the sum, across active products, of products.stock multiplied by the effective current unit price. The effective current unit price is products.sale_price when sale_price is present; otherwise it is products.price. The canonical calculation is stock multiplied by COALESCE(sale_price, price).

Current sale price

A non-null `sale_price` is the effective current price for the inventory-value metric. This controlled metric does not require `sale_price` to be lower than `price`. Do not replace missing `sale_price` with zero; use the regular price.

Category relationships

A product belongs to its direct category through `products.category_id` equals `categories.category_id`. A category may reference a parent through `categories.parent_category_id`. Use the direct category unless the request explicitly asks for a parent-category rollup. Do not infer categories from names or SKU values.

Recognized product sales by category are governed by the Sales and Order Analytics Policy. They use `order_items.quantity` multiplied by `order_items.unit_price` for completed and shipped orders; they do not use current product price or current stock.

Coupon validity

A coupon is active and valid on the frozen reporting reference date only when `coupons.active` is true, `valid_from` is null or no later than the reference time, and `valid_to` is null or no earlier than the reference date. The frozen reference date is 31 August 2026 UTC. Return the coupon code, discount type, discount value, `valid_from`, and `valid_to` when requested.

Coupon validity is independent from whether an order used the coupon. Order counts by coupon include coupons with zero related orders through a left join. Recognized discount amounts by coupon apply the completed-or-shipped order filter defined in the Sales and Order Analytics Policy.

Disambiguation rules

- Product-status counts include every product status and do not filter stock.
- The under-ten inventory report requires both active status and stock below 10.
- Active inventory value uses current product stock and `COALESCE(sale_price, price)`.
- Historical product sales use order-item `unit_price`, not the current catalog price.
- Coupon validity must not be added to coupon usage or coupon discount reports unless validity is requested.
- Do not add joins or filters that are unrelated to the requested metric.

Research governance

This is a synthetic controlled research policy, not an audited production merchandising policy. Any change to the schema, status vocabulary, stock threshold, pricing rule, category relationship, coupon date, or reporting reference date requires a new version and a synchronized benchmark update. Benchmark questions and expected result values are intentionally excluded from this retrieval document.